



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 7 • STANDARD 6.EE.6

Write Equations

Lesson 7-1 • Teacher Copy

Name: Type your name

Date: Today's date

Class: Class period

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$$x + 2 = 7$$

balanced with =

$x + 5 = 12$ means 'some number plus 5 equals 12'

Equation

Write the definition:

x

stands for a number

In $n + 3 = 10$, the letter n stands for the unknown number ($n = 7$)

Variable

Write the definition:

$$x + 2 = 7$$

balanced with =

$7 + 3 = 10$ — both sides equal 10

Equal sign

Write the definition:

$$3x + 5$$

no equal sign

$2x + 5$ is an expression; it becomes an equation when you write $2x + 5 = 15$

Expression

Write the definition:

$$2x + 5$$

constant = 5 (fixed)

*In $x + 7 = 12$, the numbers 7 and 12 are constants;
only x can change.*

Constant

Write the definition:



number

*In $n + 5 = 12$, n is the unknown; you solve to find n
 $= 7$.*

Unknown

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can write an equation to represent a real-world situation.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Equation

Variable

Equal sign

Expression

Constant

Unknown



Tap any word to see what it means and a picture.

1

A math sentence that says two amounts are equal, with an = sign, is an

2

A letter that stands for an unknown number is a .

3

The symbol = that shows two amounts are the same is the

.

4

A group of numbers and variables with no equal sign is an

.

5

A fixed number that does not change is a .

6

The value you are trying to find in an equation is the .



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: Which equation represents: 'A number plus 12 equals 30'?

A. $n + 12 = 30$

B. $n - 12 = 30$

C. $12n = 30$

D. $n / 12 = 30$

Step 1 'Plus' means addition, so the equation is $n + 12 = 30$.

 **Answer:** A. $n + 12 = 30$

 We do — together

Solve this one with your class using the same steps.

Problem: Which equation represents: 'Seven less than a number is 18'?

A. $n - 7 = 18$

B. $7 - n = 18$

C. $n + 7 = 18$

D. $n / 7 = 18$

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: Which equation represents: 'Three times a number equals 21'?

A. $3n = 21$

B. $n + 3 = 21$

C. $n - 3 = 21$

D. $n / 3 = 21$

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. The lab tech says: '5 times the number of evidence bags b equals 35.' Which equation represents this clue?

- A. $5b = 35$
- B. $b + 5 = 35$
- C. $b - 5 = 35$
- D. $b / 5 = 35$

Show your work:

2. A torn note reads: 'A suspect's case minutes m , after 9 fewer, leaves 24.' Which equation fits?

- A. $m - 9 = 24$
- B. $m + 9 = 24$
- C. $9m = 24$
- D. $m / 9 = 24$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Write a real-world situation that can be modeled by the equation $n / 5 = 9$. Explain what n represents and verify your equation makes sense.

Sentence starter: Situation: _____. In this problem, n represents _____. Check: $n = \underline{\hspace{1cm}}$ because $\underline{\hspace{1cm}} \div 5 = 9$.

Show your work:

Reflect — Exit Ticket

Which equation represents: 'A number divided by 6 equals 9'?

- A. $n / 6 = 9$
- B. $6n = 9$
- C. $n - 6 = 9$
- D. $n + 6 = 9$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Equation
2. **Notes 2:** Variable
3. **Notes 3:** Equal sign
4. **Notes 4:** Expression
5. **Notes 5:** Constant
6. **Notes 6:** Unknown
7. **We Try (notes):** A. $n - 7 = 18$ — *'Seven less than a number' means start with the number and subtract 7: $n - 7 = 18$.*
8. **You Try (notes):** A. $3n = 21$ — *'Three times' means multiply by 3, so the equation is $3n = 21$.*
9. **Try It 1:** A. $5b = 35$ — *'5 times the number' means 5 times b, so $5b = 35$. Check: $5 \times 7 = 35$.*
10. **Try It 2:** A. $m - 9 = 24$ — *'9 fewer' means subtract 9, so $m - 9 = 24$. Check: $33 - 9 = 24$.*
11. **Exit Ticket:** A. $n / 6 = 9$ — *'Divided by 6' means division: $n / 6 = 9$. Check: $54 / 6 = 9$ ✓*